

Combinational Circuits

Digital Electronics

Unit - 2

Syllabus यूरी यरी ?

↳ Technically = 10 units

March $\Rightarrow$ 30 days

Feb $\Rightarrow$ 10 days

Apr $\Rightarrow$ 30 days

70 days

200
4
50

Per Day $\Rightarrow 210 / 70 = \underline{\underline{3 \text{ hours}}}$

for 10 units $\Rightarrow \underline{\underline{210 \text{ Lectures max}}}$

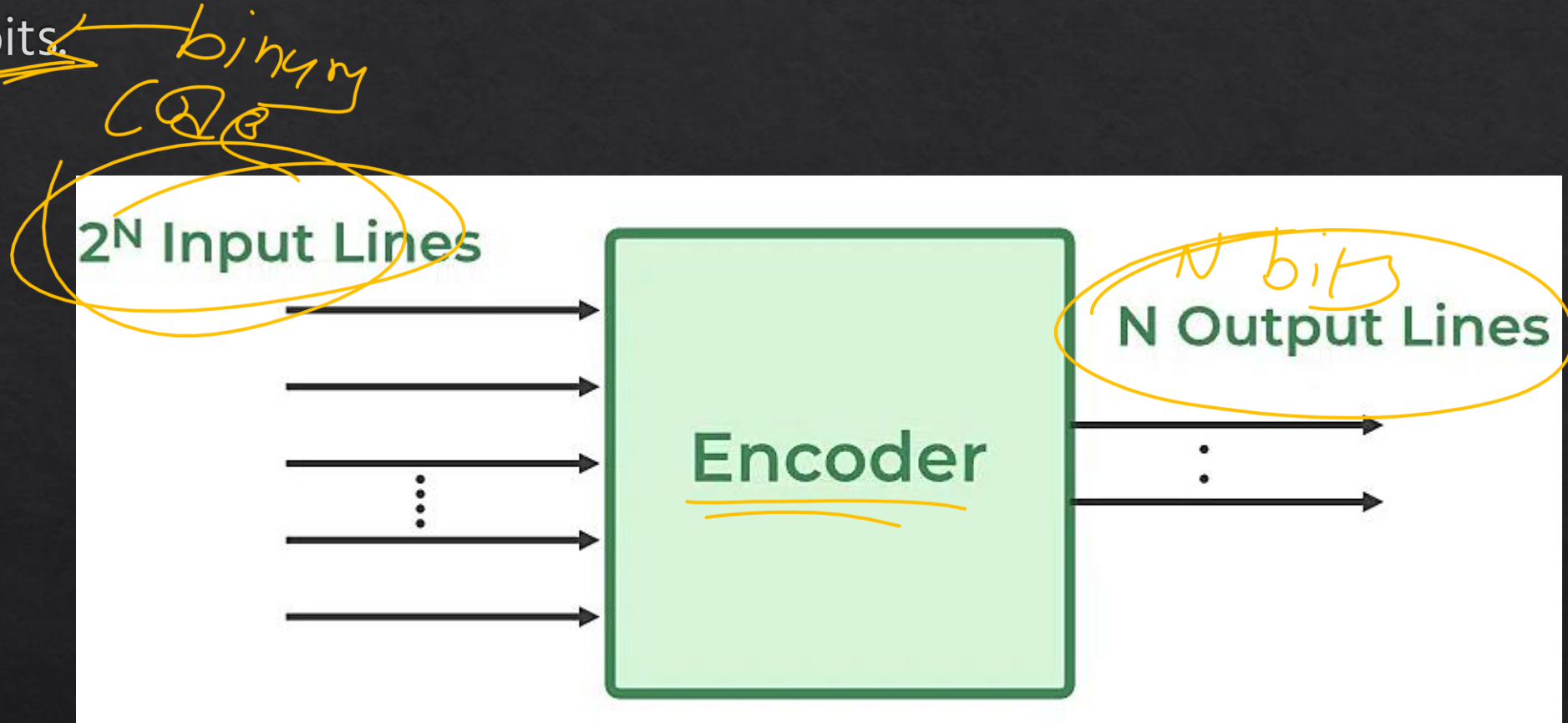
$\rightarrow$ 23rd feb $\Rightarrow$ Daily (2+2 = 4 hours)

Encoders

- ◆ An encoder in digital electronics is a combinational circuit that converts a set of input signals into a binary code.
- ◆ In other words, an encoder is a digital device that converts a piece of information represented in the form of decimal digits and alphabetic characters into a coded form like binary representation.
- ◆ Encoders are used to compress data or convert it into a format that can be easily transmitted, stored, or processed.

→ APIA

- ♦ An encoder can have maximum of 2^n input lines whereas n output lines.
- ♦ Hence, an encoder encodes information represented by 2^n input lines with n bits.



Types of Encoders

◆ Encoders can be classified based on the number of input and output lines. The most common types of encoders include:

✓ ◆ 4 to 2 Encoder

→ $(2^n) \times (n)$

$4 \div 2 \Rightarrow 2^2 : 2$

◆ 8 to 3 Encoder (Octal Encoder)

→ Octal to Binary

✓ ◆ Decimal to BCD Encoder

⇒ Decimal to (8421) (4 bit binary) ← BCD

◆ Priority Encoder

interesting

◆ Each type has specific input and output arrangements that are designed for different practical applications.

8 to 3 Encoder (Octal Encoder)

◆ An 8 to 3 encoder has eight input lines and produces a three-bit binary output.

→ [0-7]

→ (421)

◆ This type of encoder is also known as an **Octal to Binary Encoder**, as it converts octal inputs into binary outputs.

Truth Table

841P

Inputs								Outputs		
I ₇	I ₆	I ₅	I ₄	I ₃	I ₂	I ₁	I ₀	Y ₂	Y ₁	Y ₀
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

0
1
2
3
4
5
6
7

Logic Development

- ◆ From the truth table, we can write the Boolean expression for the outputs of the octal to binary encoder as follows.

$$Y_0 = I_1 + I_3 + I_5 + I_7$$

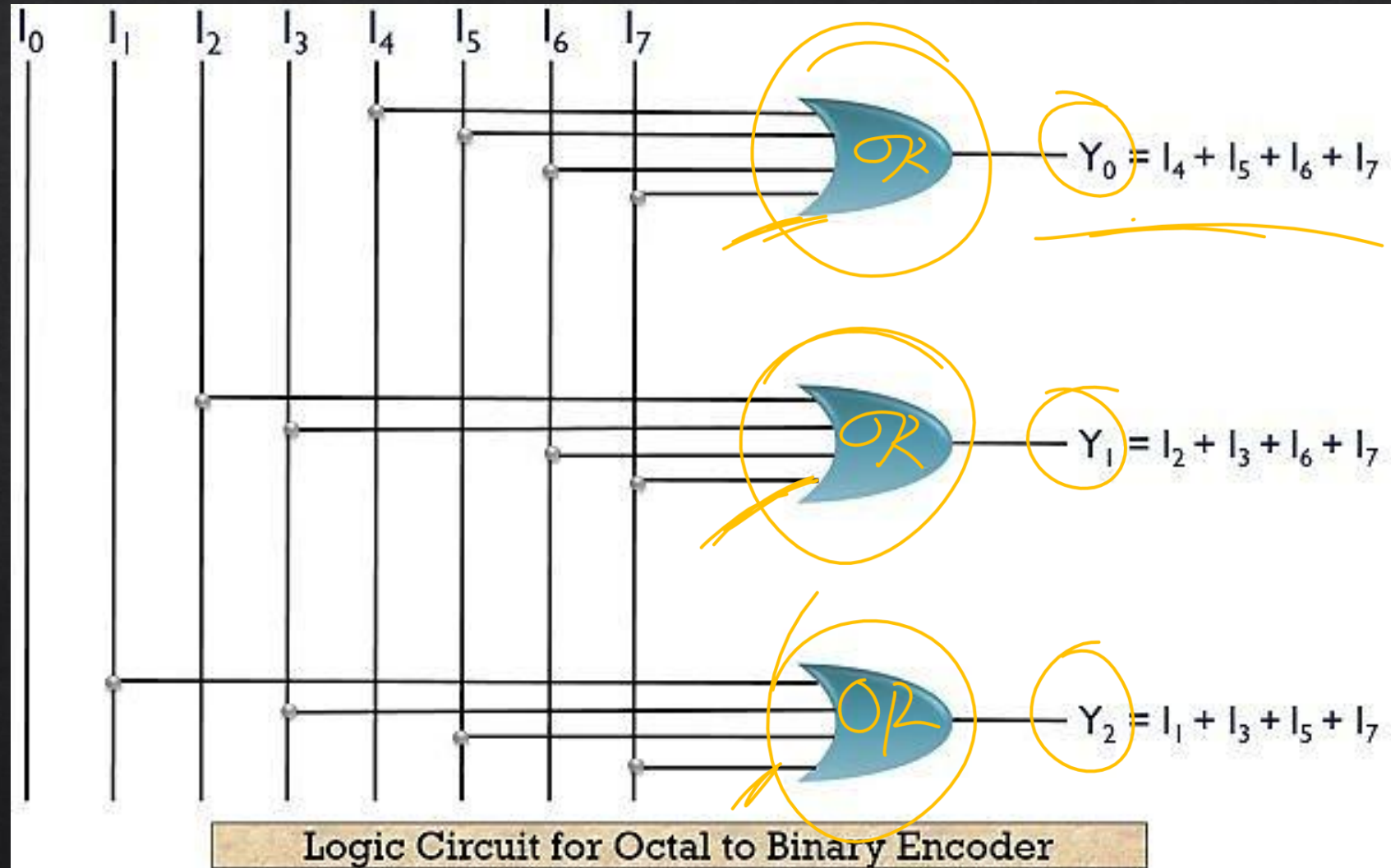
$$Y_1 = I_2 + I_3 + I_6 + I_7$$

$$Y_2 = I_4 + I_5 + I_6 + I_7$$

- ◆ From these expressions, it is clear that the implementation of an octal to binary encoder requires 3 OR gates.

MLQ

logic circuit diagram

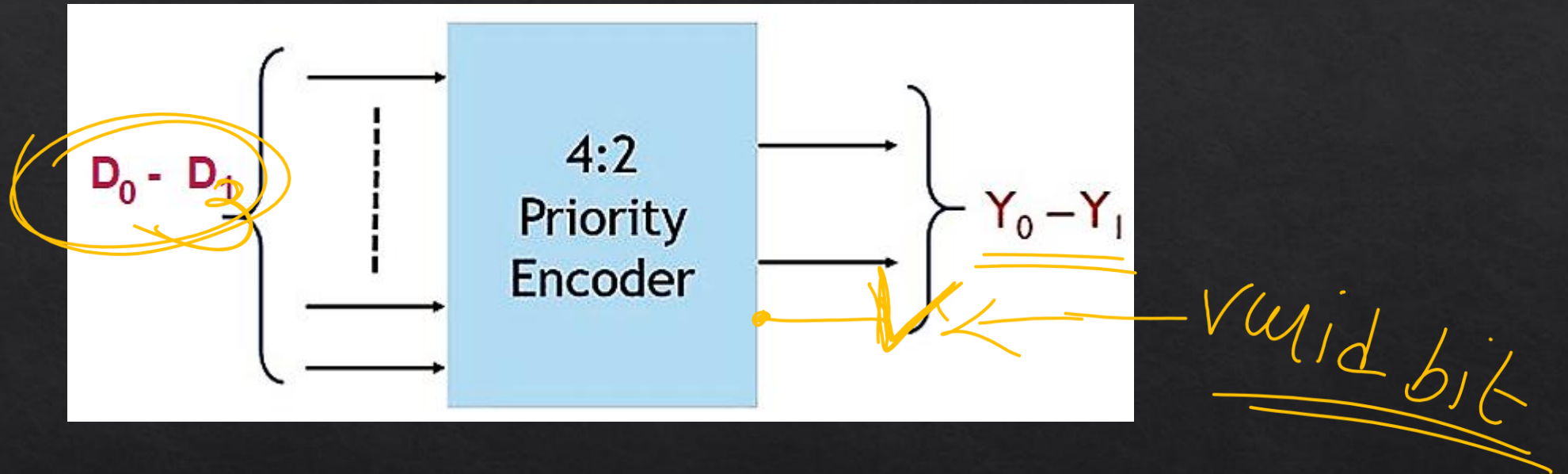


Priority Encoder

- ◆ Priority encoders are a special type of encoders that were developed to eliminate the drawback associated with normal encoders.
- ◆ As when multiple inputs are high then the encoder will not be able to correctly respond to any one of the input. As in this case, an ambiguity will get generated.
- ◆ Due to this, priority encoders were taken into consideration.
- ◆ In priority encoders, ***with movement in downward direction, priority increases***
- ◆ This means that the LSB will contain the lowest priority while the MSB has the highest priority.

4 to 2 Bit Priority Encoder

- ◆ This is also called a 4-bit priority, which contains 4 inputs and 2 output lines.



- ◆ Here, D_0 to D_3 represents the 4 input lines and Y_0 and Y_1 shows the output lines.
- ◆ Also, the priority encoder contains a 3rd output line V which is termed as a valid bit.

Decoder

Working

- ♦ If one or more inputs are 1, then the V bit will show 1 at the output.
- ♦ If all the 4 inputs are 0, there is no valid Input, thus the V bit will show 0 at the output.
- ♦ The other two outputs, i.e. Y_0 and Y_1 of the encoder are not determined when V is equal to 0.
- ♦ Therefore, when, V is equal to 0, the outputs Y_0 and Y_1 are specified as "**dont care conditions**".

- ♦ So, we can say that at any point of time, if D_3 will be high, then the circuit does not need to check any other input line.

$\rightarrow D_0, D_1, D_2, D_3$

- ♦ As among all the input data lines shown above, D_3 holds the highest priority.
- ♦ Thereby automatically the output will be generated according to the input D_3 .

Truth Table

Inputs (X = Dont care)				Outputs		
D ₀	D ₁	D ₂	D ₃	Y ₁	Y ₀	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

D₀
D₁
D₂
D₃
high

D₃ = 1

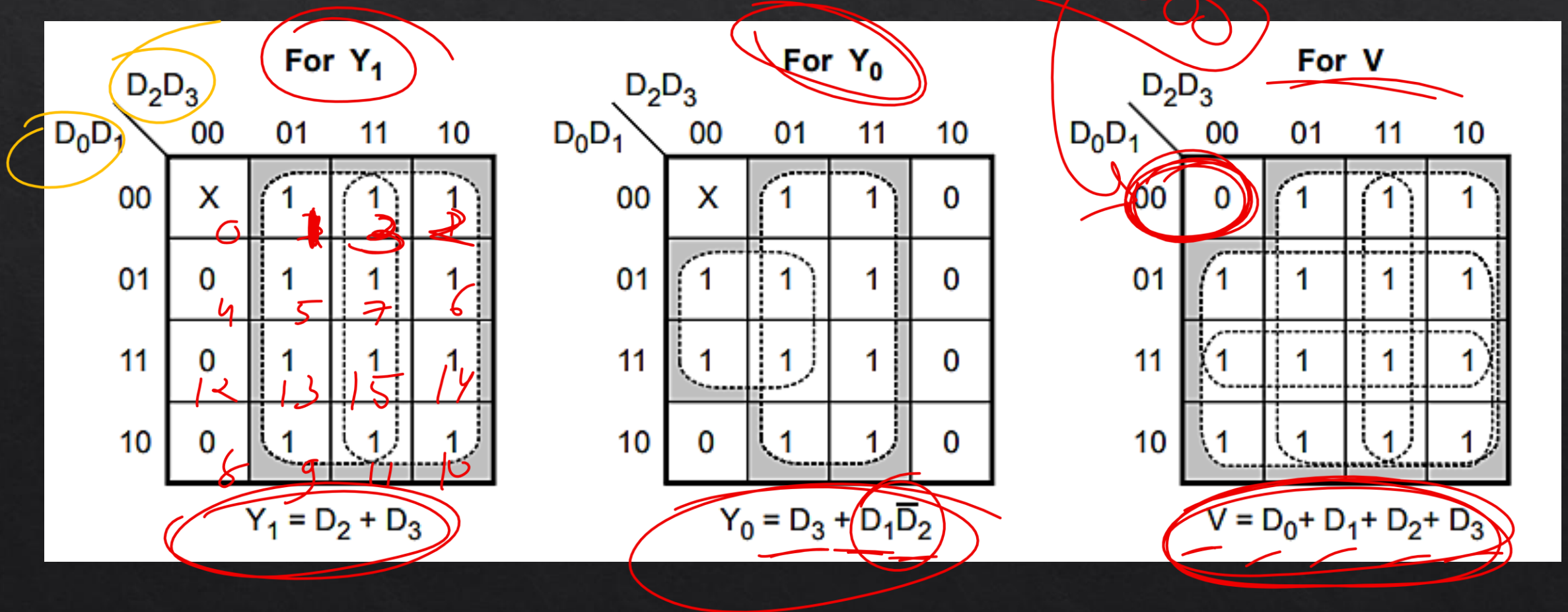
15

Applications of Octal to Binary Encoder

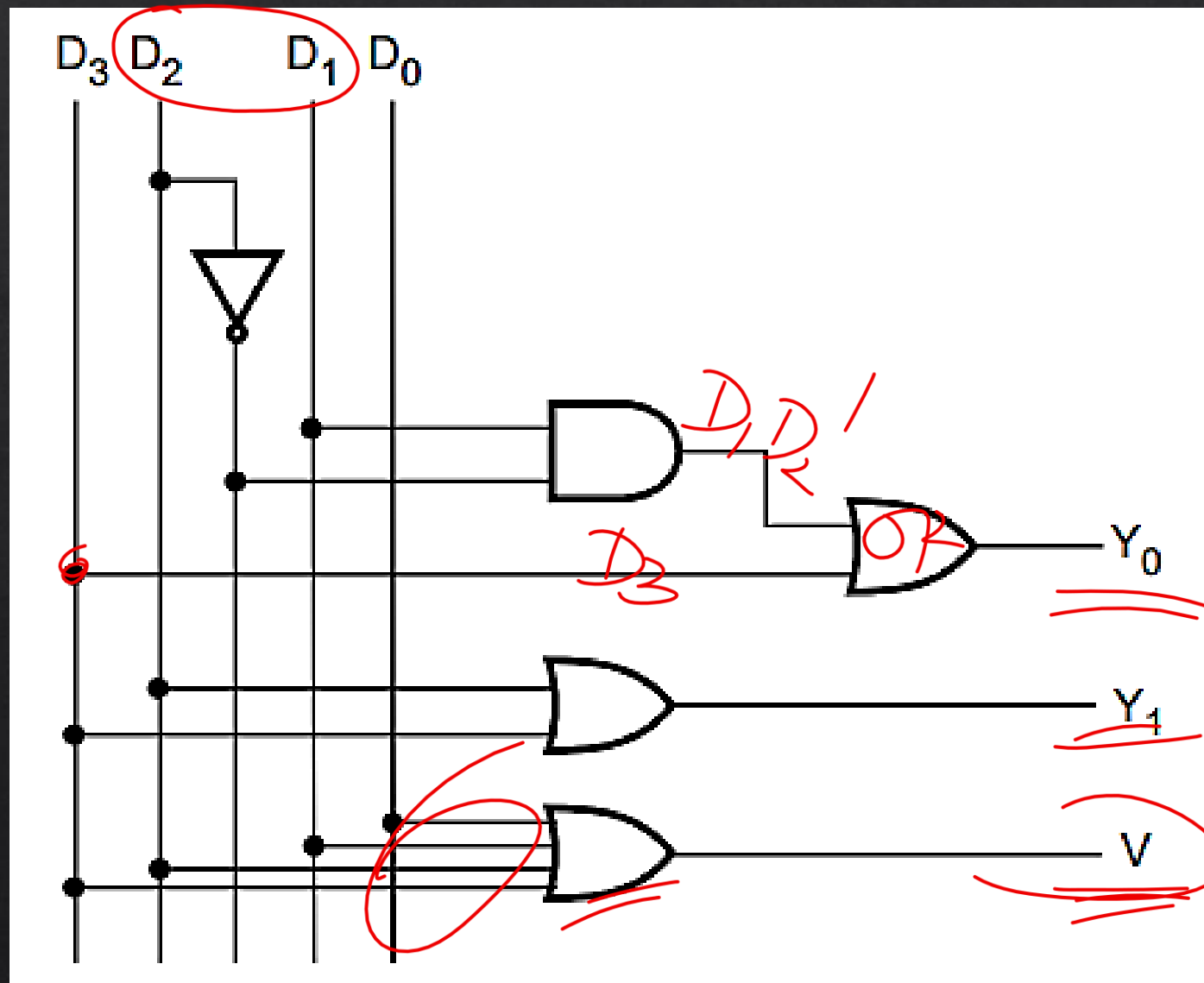
- ◆ **Multiplexers:** Converts parallel data bus to serial data stream for transmission or vice versa. $2^n \times n$
- ◆ **Communication Systems:** Encodes digital signals for efficient transmission over limited bandwidth channels.
- ◆ **Keyboards and Scanners:** Encodes key presses/scan lines into digital format. Basic info
- ◆ **Disc Drives:** Encodes data on disks using RLL, MFM or PRML codes for higher density storage.
- ◆ **Seven Segment Displays:** Uses BCD encoders to display decimal numbers. (BCD to 7-segment)
Digital
- ◆ **Barcode Readers:** Decodes barcodes scanned as varying reflectance into digital data format.
|||||

K-Map Solution

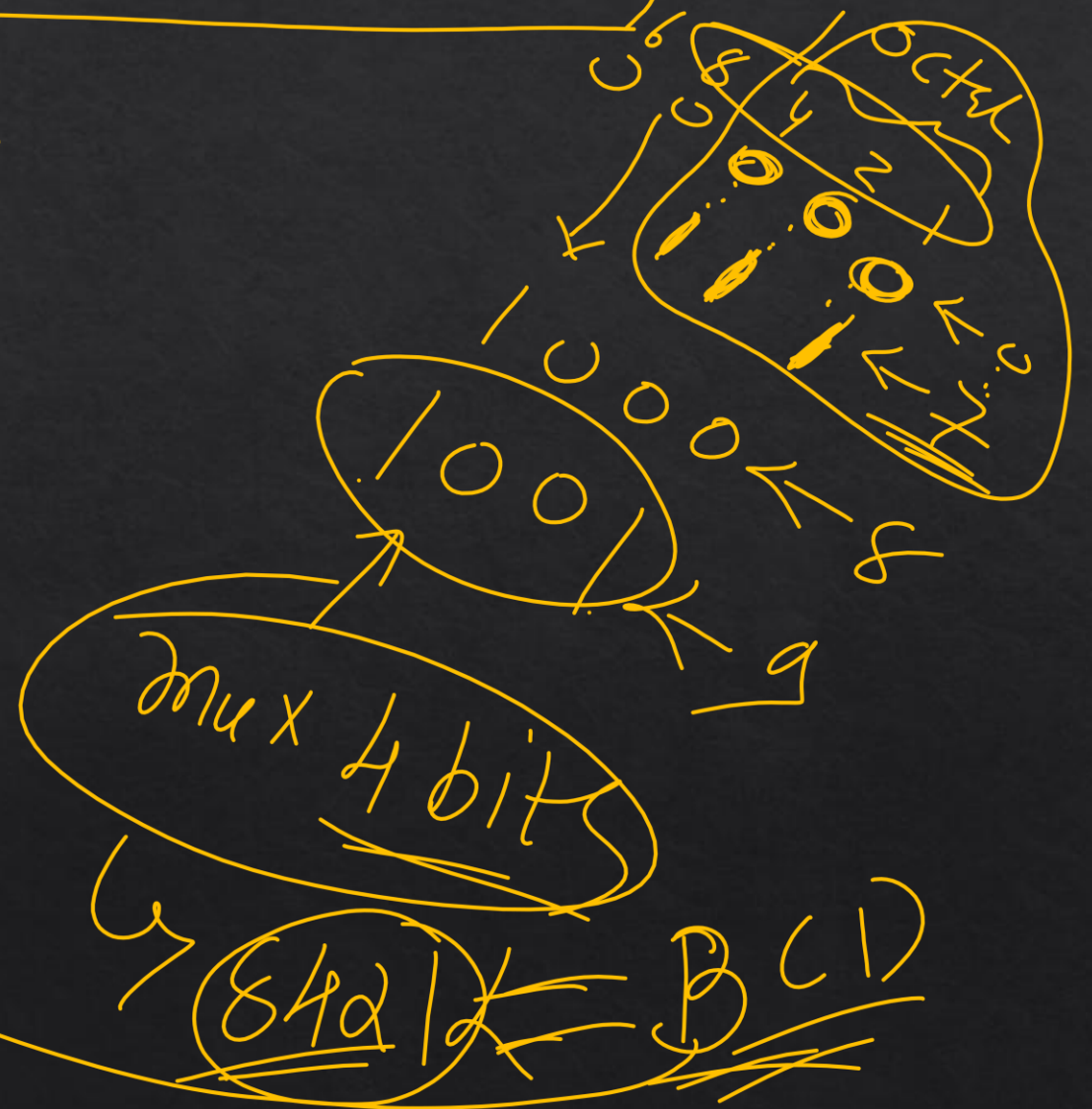
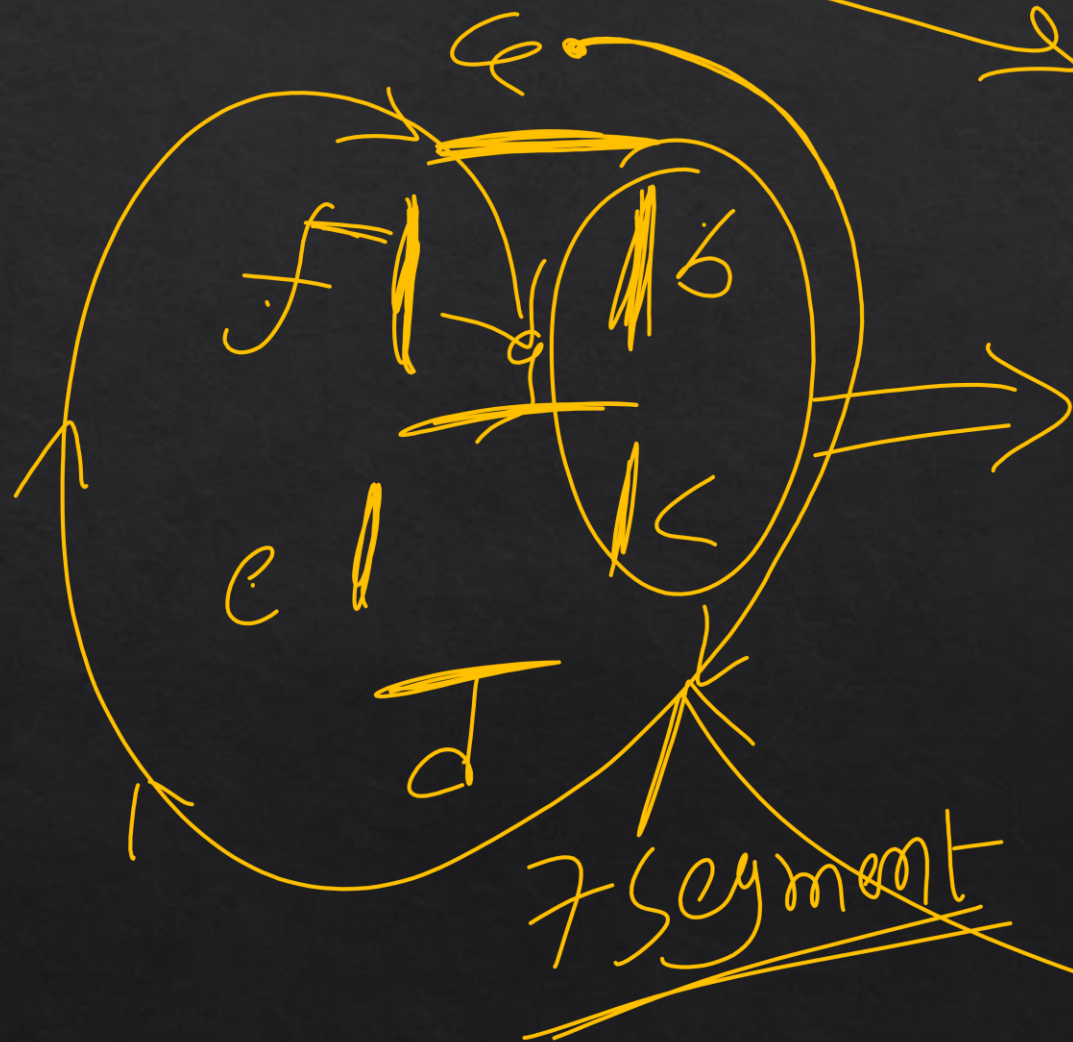
- ◇ Output expression can be told in a 4-bit encoder with the help of a K-map, as shown below:



Circuit Diagram:



7-Segment (Digital Display)

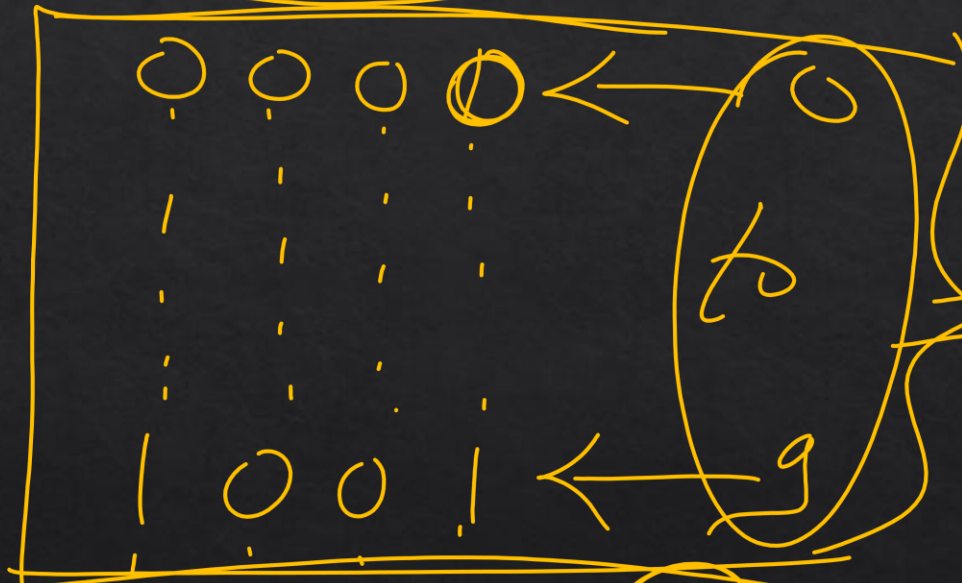


What about 10 → 15?

← 4 bit
8421

5
15

BCD



as 1 Digit
in 7-segments

are not
considered
as BCD

1001 ← 10
1111 ← 15

8x1 mux Using 2x1 selection lines



7 $\Rightarrow$ 2x1
 $\rightarrow$ mux required to Design 8x1 mux

Decoders

- ◇ A Decoder is a combinational circuit that convert binary code into a set of signals.
- ◇ It is called a decoder because it does the reverse of encoding.
- ◇ Decoders accept n input signals and produce 2^n output signals.
- ◇ Hence, the size of the decoder is $n \times 2^n$

